

Does Affirmative Action Ban Reduce Enrollment and Graduation Share of Minorities?

Sunny Xu

School of Global Policy and Strategy
University of California, San Diego

Abstract. This study evaluates the impact of Affirmative Action bans on the enrollment and graduation shares of Black and Hispanic students in undergraduate public institutions. Using a difference-in-differences approach, the main findings reveal that Affirmative Action bans lead to a modest decrease in Black enrollment shares on average at different selectivity levels, with negligible effects on graduation shares. For Hispanic students, the ban results in a 2% increase in enrollment shares among lower-end institutions, but a 2% decrease among higher-end institutions, with minimal impact on graduation shares. As the impact on graduation is less pronounced than enrollment, the study implies the possibility of the mismatch hypothesis.

Keywords: Affirmative Action Ban · Mismatch Hypothesis · Causal Inference

I Introduction

Affirmative Action remains a contentious issue in the United States. This policy aims to provide admissions advantages to underrepresented minorities (URM) during the college admissions process by accepting lower test scores and using more relaxed criteria for candidates. Despite its intentions to address racial disparities, the policy has faced criticism, with opponents arguing that it limits opportunities for white and Asian applicants and leads to reverse discrimination. In light of these ongoing debates, this study seeks to address the research question: What are the effects of Affirmative Action bans on the enrollment and graduation share of URM in 4-year public universities at the institutional level?

Understanding the impact of Affirmative Action policies is crucial to inform policymakers and ensure that efforts to reduce racial disparities achieve the desired outcomes without producing unintended consequences. This study is novel and contributes to the literature by utilizing the most updated data available and comparing the differences in the effect between enrollment and graduation share to potentially test for the mismatch hypothesis. By doing so, it aims to provide valuable insights for policymakers and contribute to a more informed discussion about Affirmative Action.

This study employs a panel of institutional-level data of fall enrollment and graduation rates from the Integrated Post-secondary Education Data System (IPEDS) in the U.S. across 50 states. Fall enrollment share and graduation share serve as outcome variables, with data available by year and race. The analysis of enrollment share covers the period from 1994 to 2021 and aims to capture the effect of Affirmative Action ban on enrollment of URM at universities of varying selectivity levels. Additionally, the study investigates graduation share between 1997 and 2021, focusing on the number of bachelor's degree completions within six years or less.

The results of this study suggest that the impact of affirmative action bans on Black enrollment and graduation share is relatively modest, while the impact on Hispanic enrollment share is more substantial. Affirmative action bans increase Hispanic enrollment share in non-AAU institutions by 2%, while decreasing enrollment share in AAU institutions by 1.8%. The effect of the ban on Hispanic graduation share is almost negligible. On the other hand, the ban has a small negative impact on Black enrollment share, and almost no effect on Black graduation share. These findings suggest a possible heterogeneity in the effects of affirmative action bans on Hispanic and Black students.

Furthermore, the results suggest that affirmative action bans may lead to a mismatch between students and universities, as there is a smaller negative impact of the bans on graduation compared to enrollment. However, it is also possible that minority students are discouraged from applying to college due to affirmative action bans. Therefore, additional research is necessary to evaluate if affirmative action bans result in fewer URM applying to colleges, particularly higher-end institutions.

The remainder of the paper is structured as follows. In Section II, I provide a literature review and highlight the limitations of previous studies. In Section III, I give more context about Affirmative

Action in the United States. Section IV gives an overview of the data in this study. Section V explains the main empirical strategy to estimate the effects of Affirmative Action bans on enrollment and graduation shares. In Section VI, I present the results of the study. Finally, Section VII concludes the paper by emphasizing the key takeaways and offering suggestions for future policy implications based on the findings.

II Literature Review

The current literature on Affirmative Action bans primarily focuses on examining application and enrollment rates, often using California as the case study example. Consequently, there is limited understanding of the overall effects of these bans and student outcomes. Card and Krueger (2005) examined the impact of bans in Texas and California, finding no evidence that highly qualified minority applicants reduced their application rates to top state institutions. Antonovics and Sander (2013) explored whether California’s Proposition 209 lowered the value that underrepresented minorities placed on attending UC schools but discovered no evidence that yield rates fell for minorities relative to other students after the implementation of Proposition 209.

While the studies by Card and Krueger (2005) and Antonovics and Sander (2013) provide valuable insights into the effect of Affirmative Action bans on application and enrollment rates, their focus on a few states. Furthermore, although Card and Krueger (2005) considered both enrollment and graduation rates, their study did not account for the potential heterogeneity of the bans in higher-end universities.

Backes (2012) is the only study that considers both enrollment and graduation rates while accounting for heterogeneity. However, it used data before 2010, and since then, three more states have passed Affirmative Action bans, providing more recent data and a longer time-frame for analysis. The existing literature does not capture the most updated information on the effect of Affirmative Action bans on minority students’ enrollment and graduation shares.

This study contributes to the existing literature by analyzing both enrollment and graduation shares across states, aiming to provide a more comprehensive understanding of the implications of Affir-

mative Action bans. The study also uses a different selectivity measure and different controls from Backes (2012). Furthermore, this research seeks to contribute to the debate on the mismatch hypothesis by comparing enrollment and graduation outcomes for minority students. Hence, this study can offer valuable insights for policymakers in evaluating the effectiveness of Affirmative Action policies and their potential consequences.

III Background

The Supreme Court addressed Affirmative Action as early as 1978 in *Regents of the University of California v. Bakke*, ruling that it was acceptable as long as schools did not establish an actual quota system for minority students (“*Regents of the University of California v. Bakke*”, n.d.). More recently, *Students for Fair Admissions Inc. v. President & Fellows of Harvard College* challenged Affirmative Action, arguing that it potentially harms Asian American applicants (“*Students for Fair Admissions v. President and Fellows of Harvard College*”, n.d.). Furthermore, there is one prevailing argument against Affirmative Action: the “Mismatch Hypothesis” (Whitt, 2018). The mismatch hypothesis suggests that by favoring less-qualified minority applicants, Affirmative Action may harm targeted students by enrolling them in colleges where they are not academically fitted. This can lead to a disparity between the student’s abilities, potentially resulting in social harm and lower outcomes later in life.

Since 1996, ten states have voted to ban the use of racial preferences in college admission: California (1998), Texas (1996), Washington (1998), Florida (1999), Michigan (2006), Nebraska (2008), Arizona (2010), New Hampshire (2012), Oklahoma (2012), and Idaho (2020). However, Texas’ ban was reversed by the 2003 *Grutter v. Bollinger* decision (“*Grutter v. Bollinger*”, n.d.).

IV Data

To estimate the impact of Affirmative Action bans on minority students, this study utilizes a panel of institutional-level data on fall enrollment and graduation rates from the Integrated Post-secondary Education Data System (IPEDS) across 50 U.S. states. The data includes 763 four-year public institutions, with data available by year and race.

Fall enrollment and graduation shares are the two primary outcome variables. To compute the share of enrollment and graduation, the number of race-specific students enrolling in a given year or completing a bachelor's degree within six years or less is divided by the total enrollment or graduates in that year, respectively.¹ The data covers the period from 1994 to 2021 for enrollment shares and from 1997 to 2021 for graduation shares.

Table 1 below provides the summary statistics for enrollment and graduation shares, as well as the share by race for each of these groups. The data is divided by state (ban states and control states²), selectivity (AAU³ and non-AAU), and sector (public and private). The table shows that there are differences in enrollment and graduation shares by race across different groups. For example, Black enrollment shares are generally higher in non-AAU and public institutions, while Hispanic enrollment shares are higher in ban states and public institutions. Asian enrollment shares are highest in AAU institutions and ban states.

Table 1. Summary Statistics

	By State			By Selectivity		By Sector	
	All States	Ban States	Control States	Non-AAU	AAU	Public	Private
Enrollment, all	833	1027	766	740	2822	1490	428
Black Enrollment Share (%)	12.8	10.2	13.2	13.1	6.1	13.5	12.4
Hispanic Enrollment Share (%)	9.1	17.2	6.35	9.1	9	10.5	8.2
Asian Enrollment Share (%)	4.2	6.25	3.5	3.8	13.8	4.8	3.8
Graduates, all	361	388	351	292	2120	787	189
Black Graduation Share (%)	11.1	8.6	11.9	11.3	4.8	12	10.7
Hispanic Graduation Share (%)	7.7	16.4	4.7	7.7	7.5	7.9	7.6
Asian Enrollment Share (%)	4.6	7.8	3.6	4.3	13.2	5.2	4.4

Notes: This table presents summary statistics for fall enrollment and graduation shares from the Integrated Post-secondary Education Data System (IPEDS) for all 50 U.S. states, covering the period from 1994 to 2021 for enrollment shares and from 1997 to 2021 for graduation shares. Enrollment and graduation rates are separated by Black, Hispanic, and Asian student groups, as well as by state, selectivity, and sector. The American Association of Universities (AAU) is defined as a group of America's leading research universities, and includes 35 public universities. Ban States refer to states that have implemented Affirmative Action bans during the sample period. All statistics presented in the table are based on institutional-level data.

¹ Although other functional forms, such as levels or logs, can be considered, Backes (2012) showed that all three yield similar results. Hence, this study uses the share of enrollment and graduation as outcome variables.

² The control states are states have not ban Affirmative Action, which serves as a control group in this study

³ The American Association of Universities (AAU) is composed of America's leading research universities, usually considered highly selective and academically competitive. Among AAU, 35 are public universities.

However, the dataset has certain limitations. A key disadvantage is the absence of information on post-college outcomes, such as the average income of individuals after graduation by race over time. Additionally, the study relies on institutional-level data rather than individual-level data.

It's important to note that the main analysis data is restricted to first-time students in four-year public universities, as public universities are required to follow state-wide Affirmative Action bans. This restriction could potentially lead minority applicants to prefer applying to private universities⁴ or universities not affected by Affirmative Action ban.⁵

A Enrollment Share by Race (1994 - 2021)

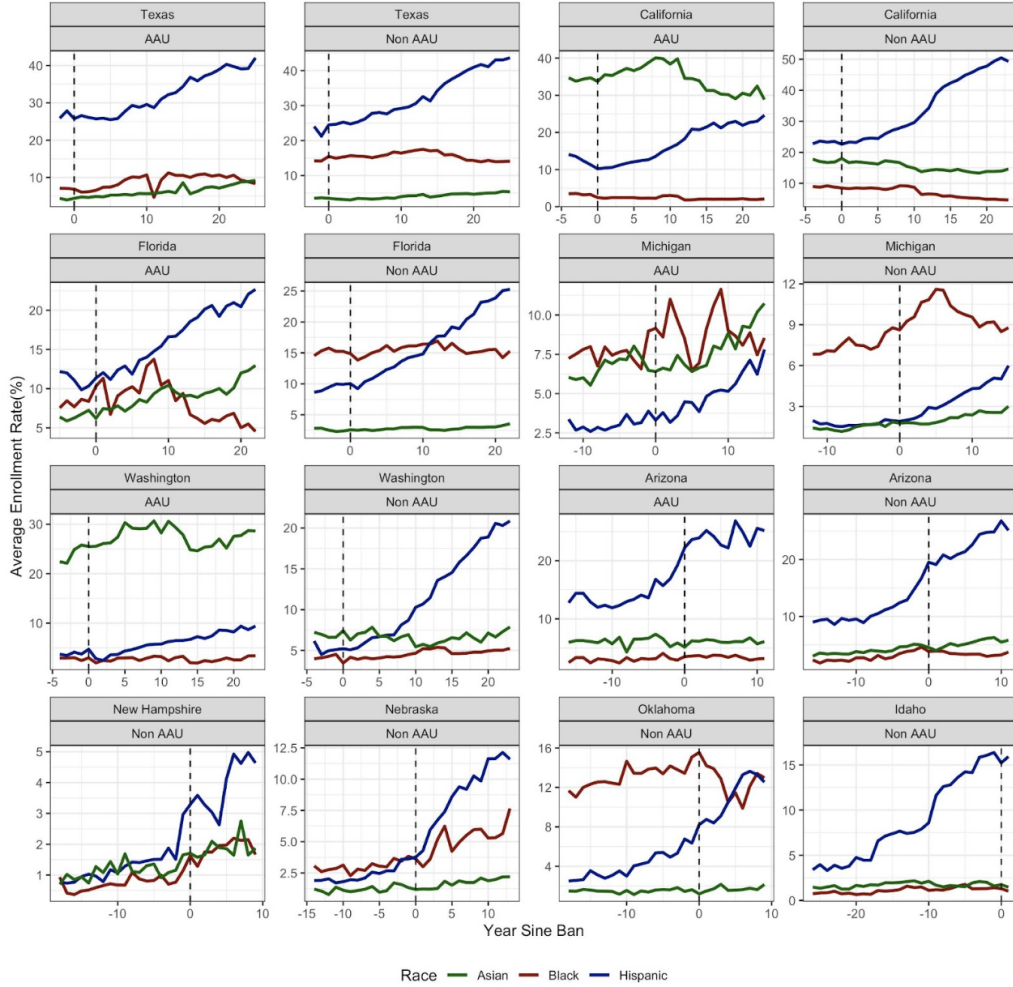
Before analyzing the impact of Affirmative Action bans on URM enrollment, I examined the trends in enrollment share of first-time undergraduate students of Black, Hispanic, and Asian groups by state and selectivity levels (AAU vs. non-AAU). Figure 1 shows the enrollment share of each group aggregated by state and selectivity levels from 1994 to 2021. Most states display similar trends in enrollment before they banned Affirmative Action, as observed in control states, with Hispanic enrollment gradually increasing and Black enrollment remaining relatively stable over time. This similarity in trends supports the parallel trends assumption, which is crucial for the identification strategy employed in this study.

Although Hispanic enrollment share increased across all states at different selectivity levels, the increase was more significant in non-AAU. For example, in Washington, in the year the state passed the ban, the average enrollment share in both AAU and non-AAU was under 10%. After the ban, Hispanic enrollment share in AAU never exceeded 10%, while Hispanic enrollment share in non-AAU increased to around 20% in 2021. Black enrollment share, on the other hand, remained relatively stable in most states at different selectivity levels. However, Florida AAU and Michigan AAU experienced some fluctuations after the ban, and the average Black enrollment in Florida AAU reduced to under 5% after 20 years of the ban. As there are differential effects between AAU and non-AAU, it's important to account for heterogeneity of selectivity levels of institutions.

⁴ Private institutions are not subject to the bans

⁵ I will briefly discuss private four-year institutions and out-of state students in the results section. For more details on the analysis, refer to the appendix.

FIGURE 1. Undergraduate Enrollment Share in Public Universities of Banned States by Race and Selectivity (1994- 2021)

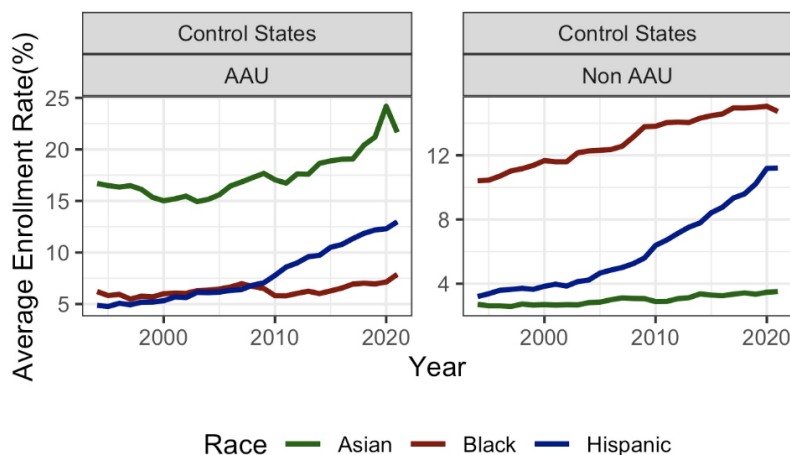


Notes: This figure shows the enrollment share of Black, Hispanic, and Asian undergraduate students in public universities of banned states aggregated by state and selectivity levels from 1994 to 2021. The data used in this figure is from the Integrated Postsecondary Education Data System (IPEDS). The Black group is represented in blue and the Hispanic group is represented in red. The Asian group is represented in green as a reference point for comparison purposes.

Figure 2 presents the enrollment share of Black, Hispanic, and Asian students in control States. Hispanic enrollment share increased over time in both AAU and non-AAU universities, while Black enrollment remained stable in AAU universities and increased in non-AAU universities. These findings highlight the importance of accounting for pre-existing time trends and variations across states.

To address this, I will include both institution and year fixed effects in my main analysis. Moreover, I removed Texas and Idaho from our analysis due to limited pre-ban data for Texas and limited post-ban data for Idaho. I'll also perform a robustness check to run the main analysis with Texas and Idaho to see how this will affect the results.

FIGURE 2. Undergraduate Enrollment Share in Public Universities of Control States by Race and Selectivity (1994- 2021)



Notes: This figure shows the enrollment share of Black, Hispanic, and Asian undergraduate students in public universities of control states aggregated by state and selectivity levels from 1994 to 2021. Control states refer to states that did not implement Affirmative Action bans during this period. The data used in this figure is from the Integrated Postsecondary Education Data System (IPEDS). The Black group is represented in blue and the Hispanic group is represented in red. The Asian group is represented in green as a reference point for comparison purposes.

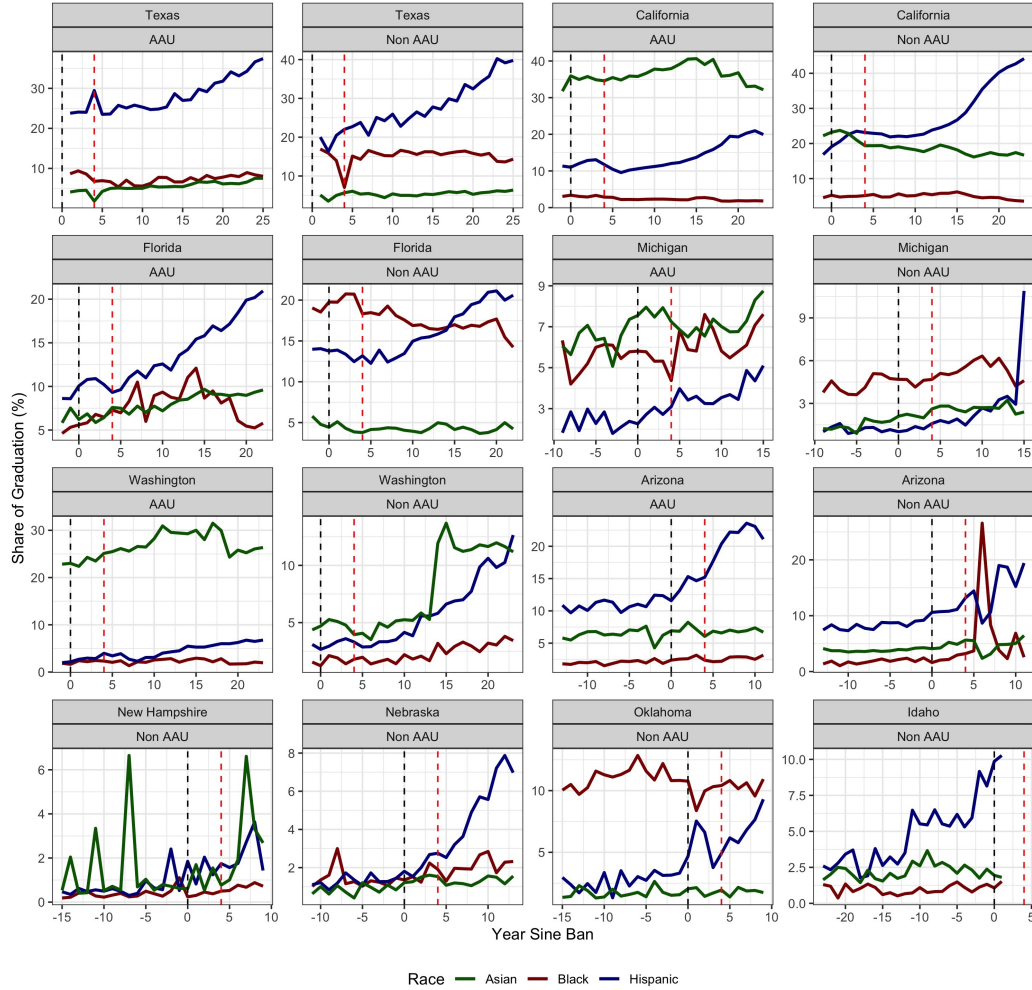
B Graduation Share Rate by Race (1997 - 2021)

As Graduation share is another variable of interest, I also looked at the trend of the graduation share of students who graduated bachelor's or equivalent degrees within 6 years or less from 1997 to 2021.

Figure 3 displays an increase in Hispanic graduation share across most states and selectivity levels, with more growth in states such as non-AAU in Texas, California, and Washington. Overall, Black graduation share demonstrate a similar pattern as enrollment share, with relatively stable rates in most states at different selectivity levels, although Florida and Michigan AAU experienced some fluctuations after the ban. However, there is a spike at 6 years after the ban in Arizona non-AAU,

which deviates from the trend of enrollment. Both indicates the heterogeneity of selectivity levels, and to account for the differential effect, an interaction term with the selectivity level will be included in the analysis.

FIGURE 3. Undergraduate Graduation Share in Public Universities of Banned States by Race and Selectivity (1997- 2021)

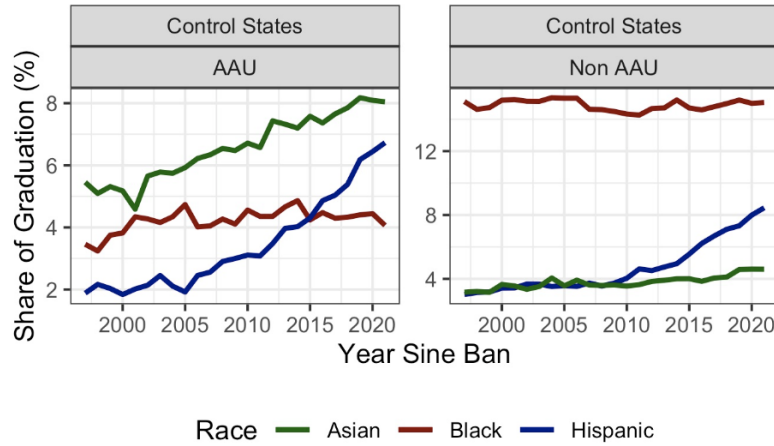


Notes: This figure shows the graduation share of Black, Hispanic, and Asian undergraduate students in public universities of banned states aggregated by state and selectivity levels from 1997 to 2021. The data used in this figure is from the Integrated Postsecondary Education Data System (IPEDS). The Black group is represented in blue and the Hispanic group is represented in red. The Asian group is represented in green as a reference point for comparison purposes.

The trend shows a more significant change 4 years after the ban. For instance, in Texas AAU and non-AAU, there was a significant decrease in Hispanic graduation share at 4 on the x-axis in AAU,

while there was a significant increase in Black graduation share in non-AAU. This implies the need to include a 4-year lead for both Black and Hispanic graduation share variables to account for the potential lag effect.

FIGURE 4. Undergraduate Graduation Share in Public Universities of Control States by Race and Selectivity (1997- 2021)



Notes: This figure shows the graduation share of Black, Hispanic, and Asian undergraduate students in public universities of control states aggregated by state and selectivity levels from 1997 to 2021. Control states refer to states that did not implement Affirmative Action bans during this period. The data used in this figure is from the Integrated Postsecondary Education Data System (IPEDS). The Black group is represented in blue and the Hispanic group is represented in red. The Asian group is represented in green as a reference point for comparison purposes.

Furthermore, Figure 4 shows the trend of graduation share in non-banned states. Both Hispanic graduation share increase over time at both AAU and non-AAU, and Black graduation share remain stable in both AAU and non-AAU. The consistent patterns across control states reinforce the parallel trends assumption, allowing for a more reliable estimation of the impact of Affirmative Action bans on graduation.

V Empirical Methodology

Estimates are based on a comparison of pre and post-policy change groups, which can be seen as a difference-in-differences approach with the ban implemented at different time. The following sections outline the analytical strategy and model specifications to examine the research question.

A Analytical Strategy

In this paper, I use a difference-in-differences approach to evaluate the effect of Affirmative Action bans on URM students' enrollment and graduation shares. This approach compares the changes in enrollment and graduation shares before and after the bans, between institutions that implemented the ban (treated units) and those that didn't (untreated units).

B Model Specifications

The estimating equation for my difference-in-differences model is as follows:

$$(1) \text{ Enrollment Share}_{ist} = \alpha_1 + \beta_1 \text{Ban}_{st} + \beta_2 \text{Ban}_{st} * \text{AAU}_i + \delta X_{st} + \gamma_i + \gamma_t + \epsilon_{ist}$$

$$(2) \text{ Graduation Share}_{ist} = \alpha_1 + \beta_1 \text{Ban}_{st} + \beta_2 \text{Ban}_{st} * \text{AAU}_i + \delta X_{st} + \gamma_i + \gamma_t + \epsilon_{ist}$$

where *Enrollment Share*_{ist} is the enrollment share (%) of Black or Hispanic for institution *i*, state *s*, and year *t*. *Graduation Share*_{ist} is the graduation share(%) of Black or Hispanic for institution *i*, state *s*, and year *t*. *Ban*_{st} is a binary variable that equals 1 if the state is affected by an Affirmative Action ban in year *t*, and 0 otherwise. *X*_{st} is a vector of control variables. γ_i is institutional fixed effects. γ_t is the year fixed effects. The error term, ϵ_{ist} , captures the unobserved factors that affect enrollment share but are not accounted for by the other variables in the model. Here, the AAU indicator has been removed from the equation as it has been absorbed by the institutional fixed effect.

I run separate regressions for Black and Hispanic students, as the enrollment and graduation share trends differ between these groups.

I define a state as being affected by an affirmative action ban if it has enacted a policy that prohibits affirmative action or if it has a court ruling that bans affirmative action. As of January 2023, nine states are still affected by Affirmative Action bans: California, Washington, Florida, Michigan, Nebraska, Arizona, Oklahoma, New Hampshire, and Idaho. Texas was affected by the affirmative action ban but later overturned it. I exclude Texas and Idaho from the dataset since there is not

enough pre and post-treatment data. Standard errors will be clustered at the state level to account for potential correlation of errors within states.

β_1 is the estimated impact of an Affirmative Action ban, representing the expected change in the outcome variable when a state is affected by a statewide ban. Following Backes (2012), I included similar controls for statewide economic characteristics to control for factors that could potentially influence enrollment and graduation shares and be correlated with Affirmative Action bans. I include the average yearly state unemployment rate, share of individuals with at least a college diploma, and average income in the state. Unlike Backes (2012), I do not include other education policy variables. One of the most common education policies as an alternative to the ban is the top-x percent program and Long (2004) showed that this program fails to replace Affirmative Action.

I hypothesize that Affirmative Action bans may have a more significant impact on enrollment and graduation rates for students in higher-end colleges compared to all colleges. Therefore, I define an institution as higher-end if it's a member of the American Association of Universities (AAU).

The fixed effects in this model are used to control for unobserved heterogeneity across institutions and time. In particular, the institutional fixed effects, γ_i , capture differences between institutions that are constant over time, such as differences in selectivity. By including institutional fixed effects in the model, we are able to control for such differences and isolate the effect of Affirmative Action bans on enrollment and graduation shares.

Similarly, the year fixed effects, γ_t , control for time-varying factors that may affect enrollment and graduation share, such as changes in government policies. By including year fixed effects in the model, we are able to account for such factors and focus on the effect of Affirmative Action bans on enrollment and graduation shares.

C Assumptions & Limitations

This study relies on a difference-in-differences approach, which comes with certain assumptions. In this section, I discuss the key assumptions underlying the analysis and the limitations to consider when interpreting the results.

Parallel Trends Assumption

To establish a causal relationship, the parallel trends assumption must be satisfied. This assumption implies that in the absence of an Affirmative Action ban, the difference in enrollment and graduation shares between institutions affected by the ban and those not affected would be constant over time.

Spillovers and Endogeneity

Spillovers between banned states and control states could occur. For example, when a state bans affirmative action, URM students might decide to move to a nearby state and apply to college there. This could indirectly affect enrollment and graduation shares of control states. I will address this concern in the results section, arguing that this is not a concern.

One possible endogeneity problem is if bans are passed in response to a rise in minority enrollment in public universities. Backes (2012) addressed this concern by rerunning the regression and dropping one state at a time.

Time-Varying Treatment Effects

Moreover, it's important to realize that treatment effects are likely not constant across time (de Chaisemartin & D'Haultfœuille, 2020) as each state banned affirmative action at different points in time. The effect can differ across states. This presents a potential problem with DID designs where different states experience the treatment at different times. Hence the estimated effect, β_1 , is a weighted average of several DID estimates.

VI Results

The primary results of the difference-in-differences estimates for Black and Hispanic enrollment share and graduation share in public universities are presented in Tables 2 and 3. The reported coefficients are expressed in percentage terms, with a coefficient of one representing a one-percentage-point change.

A Black Enrollment and Graduation Share

Panel A of Table 2 shows the results for Black enrollment share in public universities across various specifications. The first column is the result without any fixed effects or controls. It indicates that the Affirmative Action Ban decreased Black enrollment share in non-AAU institutions by 6.5% on average. However, after including institution fixed effects, the effect of the Affirmative Action Ban becomes smaller. In column 6, the estimates including both institution and year fixed effects and controls show that the Affirmative Action ban decreased Black enrollment share in non-AAU institutions by 0.5%. Compared to the average Black enrollment share (13.6%), the effect of the Affirmative Action ban in reducing Black enrollment share in non-AAU four-year public universities was relatively modest. Moreover, the coefficient of AAU variable drops. This is likely due to institution fixed effects absorbed the effect of being a member of AAU on enrollment as AAU status does not change over time. The additional effect of Affirmative Action ban on AAU is small and negative(0.02%) and insignificant, with large standard errors of 0.5 for the interaction term between the ban and AAU variable.

Estimates of the ban on Black graduation share in public universities is shown in Panel B of Table 2. The Affirmative Action ban decreased Black graduation share in non-AAU institutions by 0.16%, which is a smaller effect than the ban on Black enrollment share. The additional effect of the Affirmative Action ban on AAU institutions is positive (0.28%), but insignificant, with large standard errors of 0.8. This implies that the Affirmative Action Ban increased the Black Graduation Share by 0.1%.⁶

B Hispanic Enrollment and Graduation Share

Table 3 presents the results for Hispanic enrollment and graduation shares in public universities. Panel A shows the main difference-in-differences estimates across various specifications for Hispanic enrollment share. Unlike the negative effect of the ban on Black enrollment, the effect of the ban on Hispanic enrollment is positive and significant. The Affirmative Action ban increased Hispanic

⁶ The additional effect is calculated by using the coefficient of Ban adding the coefficient of the interaction term: $-0.158 + 0.277 = 0.119\%$

enrollment share in non-AAU institutions by 2% on average, while the additional effect of the ban on AAU institutions is negative, decreasing Hispanic enrollment share by 1.8%⁷.

Panel B of Table 3 presents the results for Hispanic graduation share in public universities, following the same regression analysis. The Affirmative Action ban has an almost negligible and insignificant effect (-0.009%) on Hispanic graduation share in non-AAU institutions after including institution fixed effects and controls. Moreover, the additional effect of the ban on Hispanic graduation share in AAU institutions is -0.192% and insignificant, meaning that the Affirmative Action ban decreases the graduation share in AAU institutions by 0.2%.⁸

In summary, the effect of the Affirmative Action ban on Black enrollment and graduation share is relatively modest, while the ban has a more substantial impact on Hispanic enrollment share in both AAU and non-AAU institutions. The ban's effect on Hispanic graduation share is less pronounced than its effect on enrollment share.

C Potential Biases and Limitations

Private Universities

One possible limitation of the analysis is private universities absorbing some students who would have otherwise attended public universities. If the number of Black and Hispanic students at the most selective private institutions increased following the ban, it could suggest that these students attended private universities instead of lower-tier public institutions. However, the estimates in Table 4 of the Appendix show insignificant changes for Black students and mixed results for Hispanic students. There does not appear to be a significant increase in minority private school enrollment. Additionally, both Black and Hispanic enrollment decreased among private AAU.

⁷ The additional effect is calculated by using the coefficient of Ban adding the coefficient of the interaction term: $2.000 - 3.818 = -1.818 \%$

⁸ The additional effect is calculated by using the coefficient of Ban adding the coefficient of the interaction term: $-0.0091 - 0.192 = -0.201 \%$

Table 2. Estimates of Affirmative Action Ban on
Black Enrollment & Graduation Share in Public Universities

	A. Black Enrollment Share(%)					
	(1)	(2)	(3)	(4)	(5)	(6)
Ban	-6.461*** (0.419)	0.262 (0.157)	-6.972*** (0.428)	-7.489*** (0.446)	-0.596*** (0.170)	-0.461** (0.173)
AAU	-9.282*** (0.678)		-9.251*** (0.678)	-7.904*** (0.670)		
Ban* AAU	4.619** (1.438)	-0.0693 (0.521)	4.569** (1.437)	2.107 (1.417)	-0.119 (0.518)	0.0216 (0.518)
Constant	15.41*** (0.181)	13.51*** (0.0419)	15.50*** (0.182)	131.8*** (5.015)	13.68*** (0.0435)	9.137*** (2.436)
Observations	17248	17248	17248	17248	17248	17248
adj. R2	0.025	0.963	0.025	0.065	0.963	0.963
Mean Enrollment Share	13.562	13.562	13.562	13.562	13.562	13.562
Institution FE	N	Y	N	N	Y	Y
Year FE	N	N	Y	N	Y	Y
Controls	N	N	N	Y	N	Y
	B. Black Graduation Share with 4-Year Lead (%)					
	(1)	(2)	(3)	(4)	(5)	(6)
Ban	-7.501*** (0.706)	0.481 (0.274)	-7.550*** (0.710)	9.439*** (0.727)	-0.0540 (0.281)	-0.158 (0.284)
AAU	-10.36*** (0.926)		-10.35*** (0.927)	-8.114*** (0.916)		
Ban* AAU	7.035*** (1.830)	0.0970 (0.822)	7.041*** (1.832)	3.616* (1.802)	-0.0238 (0.819)	0.277 (0.821)
Constant	14.57*** (0.261)	12.59*** (0.0524)	14.58*** (0.261)	184.6*** (8.418)	12.68*** (0.0532)	9.198* (3.653)
Observations	10232	10227	10232	10232	10227	10227
adj. R2	0.024	0.978	0.022	0.071	0.978	0.978
Mean Graduation Share	12.659	12.659	12.659	12.659	12.659	12.659
Institution FE	N	Y	N	N	Y	Y
Year FE	N	N	Y	N	Y	Y
Controls	N	N	N	Y	N	Y

Notes: Institution-by-year level regressions were conducted for all 763 public universities. Panel A uses enrollment shares of Black from 1994 to 2021. Black Enrollment Share(%) is measured as the number of Black first-time, undergraduate enrolled divided by the total number of students enrolled, multiplied by one hundred. Panel B uses graduation shares of Blacks from 1997 to 2021. Black Graduation Share(%) is measured as the number of Black students who graduated bachelor's or equivalent degrees within 6 years or less divided by the total number of students enrolled, multiplied by one hundred. A four-year lead of the Black Graduation Share variable was included in the analysis to account for potential lag effects of Affirmative Action bans on graduation shares. The variable Ban is an indicator that takes a value of one if the state where the institution is located is affected by an Affirmative Action ban at year t . The variable AAU is an indicator of selectivity, taking a value of one if the institution is considered a member of the American Association of Universities (AAU). The mean and standard deviation of the dependent variable are reported in the table. Standard errors clustered at the institution level are shown in parentheses. *** is significant at the 1% level, ** is significant at the 5% level, * is significant at the 10% level.

Table 3. Estimates of Affirmative Action Ban on
Hispanic Enrollment & Graduation Share in Public Universities

	A. Hispanic Enrollment Share(%)					
	(1)	(2)	(3)	(4)	(5)	(6)
Ban	12.30*** (0.238)	7.026*** (0.186)	11.54*** (0.236)	6.767*** (0.234)	2.691*** (0.153)	2.000*** (0.151)
AAU	-1.481*** (0.385)		-1.437*** (0.374)	1.091** (0.352)		
Ban* AAU	-3.725*** (0.815)	-3.059*** (0.617)	-3.791*** (0.793)	-8.364*** (0.744)	-3.056*** (0.466)	-3.818*** (0.452)
Constant	6.818*** (0.103)	7.700*** (0.0496)	6.961*** (0.100)	108.0*** (2.631)	8.524*** (0.0392)	37.01*** (2.125)
Observations	17248	17248	17248	17248	17248	17248
adj. R2	0.141	0.857	0.187	0.295	0.919	0.924
Mean Enrollment Share	8.984	8.984	8.984	8.984	8.984	8.984
Institution FE	N	Y	N	N	Y	Y
Year FE	N	N	Y	N	Y	Y
Controls	N	N	N	Y	N	Y
	B. Hispanic Graduation Share with 4-Year Lead (%)					
	(1)	(2)	(3)	(4)	(5)	(6)
Ban	9.377*** (0.272)	1.452*** (0.230)	9.203*** (0.272)	6.092*** (0.266)	-0.137 (0.216)	-0.00910 (0.218)
AAU	-1.095** (0.357)		-1.058** (0.355)	1.050** (0.335)		
Ban* AAU	-2.722*** (0.706)	-0.0693 (0.521)	-2.634*** (0.701)	-6.169*** (0.660)	0.198 (0.630)	-0.192 (0.631)
Constant	4.108*** (0.100)	5.120*** (0.0440)	4.129*** (0.0999)	102.8*** (3.084)	5.354*** (0.0409)	3.865 (2.805)
Observations	10232	10227	10232	10232	10227	10227
adj. R2	0.112	0.905	0.123	0.237	0.921	0.922
Mean Graduation Share	6.068	6.068	6.068	6.068	6.068	6.068
Institution FE	N	Y	N	N	Y	Y
Year FE	N	N	Y	N	Y	Y
Controls	N	N	N	Y	N	Y

Notes: Institution-by-year level regressions were conducted for all 763 public universities. Panel A uses enrollment share of Hispanic from 1994 to 2021. Hispanic Enrollment Share(%) is measured as the number of Hispanic first-time, undergraduate enrolled divided by the total number of students enrolled, multiplied by one hundred. Panel B uses graduation share of Hispanic from 1997 to 2021. Hispanic Graduation Share(%) is measured as the number of Hispanic students who graduated bachelor's or equivalent degrees within 6 years or less divided by the total number of students enrolled, multiplied by one hundred. A four-year lead of the Hispanic Graduation Share variable was included in the analysis to account for potential lag effects of Affirmative Action bans on graduation rates. The variable Ban is an indicator that takes a value of one if the state where the institution is located is affected by an Affirmative Action Ban at year t . The variable AAU is an indicator of selectivity, taking a value of one if the institution is considered a member of the American Association of Universities (AAU). The mean and standard deviation of the dependent variable are reported in the table. Standard errors clustered at the institution level are shown in parentheses. *** is significant at the 1% level, ** is significant at the 5% level, * is significant at the 10% level.

Out-of-State Students

Another potential bias is the possibility that minority students applied to colleges in nearby states that didn't ban Affirmative Action. Previous research by Backes (2012) find no evidence that Black and Hispanic students reacted to the bans by attending college in nearby states.

Non-reported Race

In the IPEDS data, some of the students didn't report their race. If the ban led minority students not to report their race, it could also lead to biased results. However, both Antonovics and Sander (2013) and Backes (2012) suggest that the majority of the unknown race category is due to White and Asian students not reporting their race during applications.

White and Asian Students

The study also conducted the same regressions for White and Asian students as a comparison. The results, shown in columns 3 and 4 of Table 4 in the Appendix, indicate that both White and Asian students experienced a decrease in enrollment among non-AAU and an increase in enrollment among AAU. Given the mean enrollment shares for White and Asian students, the ban's effect appears to be more significant for Asian students.

Overall, while some factors cannot be directly tested for, previous research provides additional context and support for the study's findings.

D Robustness Checks

As mentioned in the Data section, a robustness check was performed to include Texas and Idaho in the analysis to assess how the results changed. The results of this robustness check are presented in Table 5 in the Appendix.

Table 5 displays the estimates for Black and Hispanic enrollment and graduation shares with a 4-year lead. The table is divided into four columns: columns 1 and 2 show the results for Black students, and columns 3 and 4 show the results for Hispanic students. The robustness check results, which

include Texas and Idaho, demonstrate that the main findings of the analysis remain consistent with the initial results. The affirmative action bans have a modest negative impact on Black enrollment and a positive impact on Hispanic enrollment. Furthermore, there is no significant differential effect of the ban on AAU for Black students. For Hispanic enrollment, there is a significant decrease in Hispanic enrollment among AAU, consistent with the main findings. The most notable difference in the robustness check is a more pronounced negative effect on Hispanic graduation with a 4-year lead among non-AAU institutions. Overall, the robustness check confirms the main findings of the study.

VII Discussion

This study employs a difference-in-differences approach to answer the primary question of interest: Does affirmative action decrease enrollment and graduation shares of Black and Hispanic students?

FIGURE 5. Impact of Affirmative Action Bans on Enrollment and Graduation Shares of Black and Hispanic Students

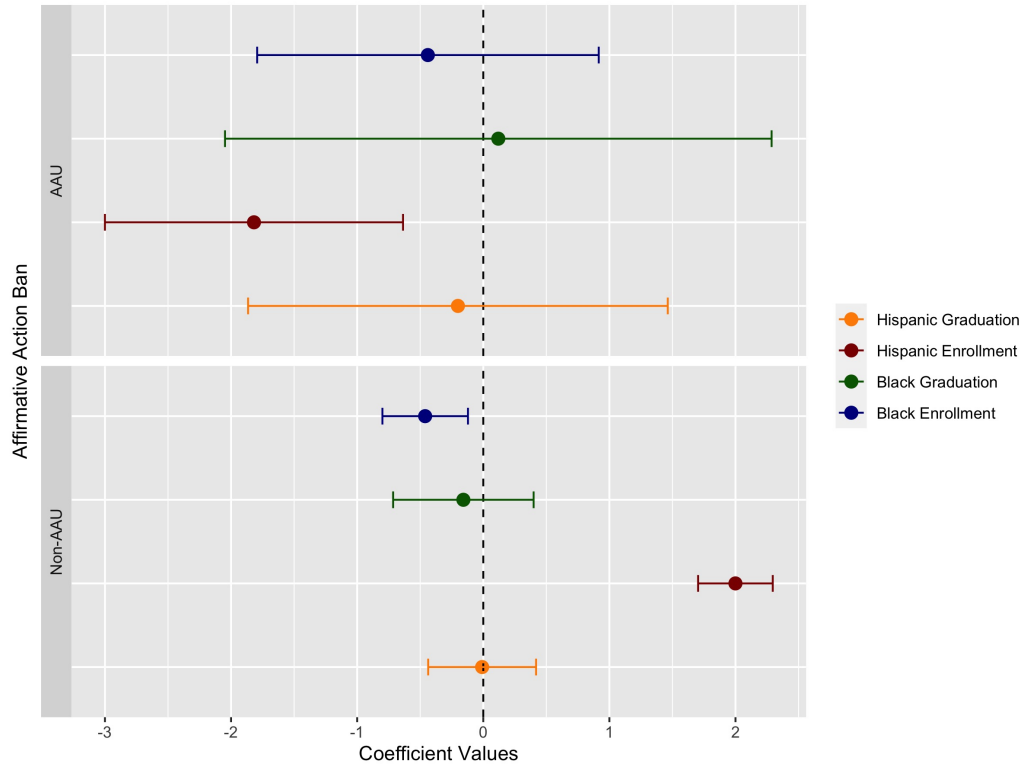


Figure 5 is a visual summary of the coefficients of the ban, allowing for a clear comparison of the effects on enrollment and graduation by race and selectivity. The main result indicates that Affirmative Action bans decrease Black enrollment shares by around 0.5% for both non-AAU and AAU institutions. The impact of Affirmative Action bans on Black graduation is minimal and almost negligible. For Hispanic students, the effects of the bans differ. Affirmative Action bans lead to an average increase of 2% in enrollment shares among non-AAU, while there is a 2% decrease in enrollment shares among AAU. Interestingly, the impact of Affirmative Action bans on Hispanic graduation is similar to that of Black students—very small and almost negligible. There seems to be heterogeneity in the effects of affirmative action bans on Hispanic students, with enrollment increasing for non-AAU but decreases for AAU.

To test the mismatch hypothesis, this study compares the effects of the bans on enrollment and graduation. If the mismatch hypothesis holds true, students are placed in institutions where they are more suitable, I would expect to see little change in graduation and potentially an increase in AAU. As shown in Figure 5, there is a smaller negative impact of the bans on graduation compared to enrollment. This finding could imply that the mismatch hypothesis might hold true.

However, another possibility is that minority students are discouraged from applying due to Affirmative Action bans. After a ban is in place, minority students who could enroll but not graduate might decide not to apply. Hence, further research is needed to evaluate if affirmative action bans lead to fewer URM applying to colleges, specifically higher-end universities.

The findings of this study have important implications for understanding the effects of affirmative action policies on minority students. By highlighting the nuances in the impact of the bans on enrollment and graduation shares, this study contributes valuable insights to the ongoing debate on the effectiveness of Affirmative Action. Policymakers should consider the differential effects of affirmative action bans on Black and Hispanic students. The study reveals that while Black enrollment shares decrease under the bans, Hispanic enrollment shows a more complex pattern, with increases in non-AAU institutions and decreases in AAU institutions. Policies should be tailored to address these varying effects on different minority groups. Policymakers should recognize the possibility of

the mismatch hypothesis and explore other ways to promote diversity and inclusion in higher education while ensuring that students are placed in institutions where they have the best chance of success.

Future research should explore the application behavior of URM in response to these bans. It can help policymakers develop targeted policies to better support minority students.

A Appendix

Table 4. Estimates of Affirmative Action with Different Enrollment Shares

	Enrollment Share(%)			
	Private Universities		Public Universities	
	Black	Hispanic	White	Asian
Ban	0.0876 (0.208)	0.967*** (0.166)	-0.989*** (0.257)	-0.457*** (0.0961)
AAU				
Ban* AAU	-1.782 (1.387)	-2.740* (1.105)	0.266 (0.769)	1.128*** (-0.288)
Constant	-0.226 (2.953)	34.19*** (2.354)	35.84*** (3.614)	5.629*** (1.353)
Observations	29232	29232	17248	17248
adj. R2	0.912	0.818	0.95	0.926
Mean Enrollment Share	12.167	7.517	62.971	4.867
Institution FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Notes: Institution-by-year level regressions were conducted for all 2010 private universities and 763 public universities. Black/Hispanic/White/Asian Enrollment Share(%) is measured as the number of Black/Hispanic/White/Asian first-time, undergraduate enrolled divided by the total number of students enrolled, multiplied by one hundred. The variable Ban is an indicator that takes a value of one if the state where the institution is located is affected by an Affirmative Action Ban at year t . The variable AAU is an indicator of selectivity, taking a value of one if the institution is considered a member of the American Association of Universities (AAU). The mean and standard deviation of the dependent variable are reported in the table. Standard errors clustered at the institution level are shown in parentheses.. *** is significant at the 1% level, ** is significant at the 5% level, * is significant at the 10% level.

Table 5. Robustness Check

	Black		Hispanic	
	Enrollment Share (%)	Graduation Share (%)	Enrollment Share (%)	Graduation Share (%)
	with 4-year lead		with 4-year lead	
Ban	-0.342* (0.164)	-0.176 (0.274)	2.098*** (0.152)	-0.207 (0.222)
AAU				
Ban* AAU	0.151 (0.483)	0.264 (0.810)	-3.857*** (0.449)	-0.198 (0.655)
Constant	10.73*** (2.376)	8.233* (3.509)	33.83*** (2.208)	0.276 (2.834)
Observations	18620	10895	18620	10895
adj. R2	0.963	0.979	0.939	0.946
Mean Enrollment Share	13.544	12.644	10.520	7.280
Institution FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y

Notes: See Notes for Table 2 and Table 3

References

- Antonovics, K. L., & Sander, R. H. (2013). Affirmative action bans and the “chilling effect”. *American Law and Economics Review*, 15(1), 252–299.
- Backes, B. (2012). Do affirmative action bans lower minority college enrollment and attainment? evidence from statewide bans. *The Journal of Human Resources*, 47(2), 435–455.
- Card, D., & Krueger, A. B. (2005). Would the elimination of affirmative action affect highly qualified minority applicants? evidence from california and texas. *Industrial and Labor Relations Review*, 58(3), 416–34.
- de Chaisemartin, C., & D’Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964–96.
- Grutter v. bollinger [Retrieved March 21, 2023, from <https://www.oyez.org/cases/2002/02-241>]. (n.d.).
- Long, M. C. (2004). Race and college admissions: An alternative to affirmative action? *Review of Economics and Statistics*, 86(4), 1020–33.
- Regents of the university of california v. bakke [Retrieved March 21, 2023, from <https://www.oyez.org/cases/1979/76-811>]. (n.d.).
- Students for fair admissions v. president and fellows of harvard college [Retrieved March 21, 2023, from <https://www.oyez.org/cases/2022/20-1199>]. (n.d.).
- Whitt, W. F. (2018). *Regarding the mismatch hypothesis and stereotype threat in the debate over affirmative action* [Student paper, EC970, Department of Economics, Harvard University].